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Faculty of Computers, Informatics, and Microelectronics
Department of Software Engineering
Software Engineering Study Programme

PROJECT PROPOSAL

B2B Web Platform for Wholesale Order Management and Optimisation of Agricultural Product Distribution Flows

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**By signing the cover page of this report, the student confirms their active involvement in both the development of the project and the writing of the report.*

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence
B2B – Business-to-Business, trade relations between companies
HoReCa – Hotels, Restaurants and Cafés, the food service sector
MVP – Minimum Viable Product
SMART – Specific, Measurable, Achievable, Relevant, Time-bound
SUS – System Usability Scale
SWOT – Strengths, Weaknesses, Opportunities, Threats
UVP – Unique Value Proposition

1 INTRODUCTION

AgroHub is a B2B web platform through which agricultural producers in the Republic of Moldova sell their produce wholesale directly to distributors, shops and HoReCa businesses. Today most of these deals are arranged by phone, in messenger groups or at wholesale markets, without a shared record of the available stock, the orders and the agreements. The platform brings these steps into one place: offers with a real stock, orders with a clear status, a conversation for every order and verified participants.

This proposal turns the general idea of the application into a justified project. Chapter 2 presents the market research: similar solutions, the needs of the users, the target user profiles and a simplified SWOT analysis. Chapter 3 refines the purpose of the project and sets two SMART objectives. Chapter 4 describes the target audience, the key problems addressed by the application and its unique value proposition. Chapter 5 summarises why the project answers a real need. A working version of the platform is publicly available at https://agrohub-efh.pages.dev/.

2 MARKET RESEARCH AND NEEDS ANALYSIS

The purpose of the research was to check whether the idea responds to a real need and to understand the users before fixing the objectives. Agriculture and food processing account for more than 18 percent of the GDP of Moldova and for about 45 percent of its exports [1], and small farms produce more than 62 percent of the agricultural output [2]. These producers rarely have a sales department: they sell through personal contacts and intermediaries, which is exactly the part of the trade the platform digitalises.

Three activities were carried out: an analysis of similar systems, the preparation of a short survey and interviews with potential users, and the definition of the target user profiles. Their results are summarised in a simplified SWOT analysis at the end of the chapter.

2.1 Similar solutions on the market

Five existing systems were analysed. Three are international platforms that digitalise trade in food or agricultural products, and two are Moldovan platforms that the target users already know. Table 2.1 gives the general information about them [3], [4], [5], [6], [7].

Table 2.1 – General information on the analysed systems

System	Producer, country, year	Business model	Main functions
Tridge	Tridge Co., South Korea, 2015	Service fees for sourcing and fulfilment, paid data subscriptions	International sourcing of agri-food products, market prices and trade data
Choco	Choco, Germany, 2018	Free for restaurants, paid tools for suppliers	Ordering from suppliers, order history, order processing for distributors
GrubMarket	GrubMarket Inc., USA, 2014	Margin on distributed products, SaaS licences (WholesaleWare)	B2B food marketplace, inventory and order management, logistics
Agromag.md	Agrobiznes, Moldova, 2022	Commission or fees paid by supplier companies	Marketplace of agricultural inputs and equipment from verified suppliers
999.md	Simpals, Moldova, 1999	Free basic ads, paid promotion of ads	General classified ads, including agricultural products

The systems differ mostly in their target market and in who pays for the service. The international platforms are financed by fees, licences or a margin on the goods, while the local ones are free or paid by supplier companies.

Table 2.2 sums up the strengths and weaknesses of each system from the point of view of a Moldovan producer who wants to sell a few tonnes of produce to a buyer in the same region.

Table 2.2 – Strengths and weaknesses of the analysed systems

System	Strengths	Weaknesses
Tridge	Global reach, market prices and trade data, support for quality checks and logistics	Built for international deals of container size; not suited to local lots of a few tonnes
Choco	Simple, structured ordering that replaces calls and voice messages; free for restaurants	Digitalises existing relationships, does not help to find new producers; not available in Moldova
GrubMarket	Real-time stock and order management for food distribution	Acts largely as a distributor itself; software made for medium and large companies in the United States
Agromag.md	Local, Romanian interface, verified suppliers	Sells seeds, fertilisers and equipment to farmers; does not help farmers sell their harvest
999.md	Very large audience, free basic ads, already used by most farmers	No stock, no orders and no verification; the deal continues by phone

None of the systems combines all the elements needed by small and medium producers in Moldova: a local and free catalogue of farm produce, a real stock on every listing, orders with a status and a conversation attached to them, and verified participants. This combination is the niche of the proposed platform.

2.2 Preliminary feedback from potential users

To compare the conclusions of the analysis with the experience of real people, a short questionnaire and semi-structured interviews of 15–20 minutes were prepared for three types of potential users: a vegetable producer, a distributor and a buyer from the HoReCa or retail segment. The questions cover how the respondents sell or buy today, what takes them the most time and what they would expect from an online platform.

At the time of writing, the interviews are scheduled but have not yet taken place. The expected answers of each respondent profile were therefore formulated as working hypotheses, derived from the analysis of the field. Table 2.3 presents them; they are not interview results and will be confirmed or corrected by the real answers.

Table 2.3 – Working hypotheses about the needs of each respondent profile

Respondent profile	Expected current practice	Expected main difficulties	Expected needs from the platform
R1, vegetable producer (10–30 ha)	Sells most of the harvest by phone to a few regular buyers, the rest through resellers or ads on 999.md	Too much time spent on calls in the harvest peak, last-minute cancellations, pressure on price from resellers	Publishing an offer from the phone in a few minutes, seeing all orders in one place, notifications
R2, distributor	Orders weekly from 10–15 producers by phone and Viber, keeps the orders in an Excel file	Does not know the real availability before calling, has to call several producers to complete one order	Search by product and region, real stock on each listing, order history, reviews of suppliers
R3, HoReCa or retail buyer	Buys from one or two distributors and sometimes directly from farmers at the market	Unstable quality and availability of fresh produce, few alternatives when a supplier fails	Quick access to alternative suppliers, simple ordering, information about the producer

The hypotheses were reduced to five statements that can be checked in the interviews. A statement is considered confirmed if at least two of the three respondents support it:

- H1 – most wholesale deals of the respondents are arranged by phone or in messengers;
- H2 – the respondents have faced at least once per season a cancelled order or a lot promised to two buyers;
- H3 – the lack of trust is the main reason for not working with new partners;
- H4 – the respondents would publish offers or order online if the operation takes only a few minutes from a phone;
- H5 – buyers value up-to-date stock information at least as much as a lower price.

The hypotheses that are not confirmed will change the requirements of the system, and the same respondents are planned to test the platform during the pilot operation described in Chapter 3.

2.3 Target user profiles

The two sides of a deal were described with the help of personas: fictional but realistic profiles that bring together the typical goals, habits and frustrations of a group of users [8]. The first persona, the producer, is shown in Table 2.4.

Table 2.4 – Persona 1: the seller (agricultural producer)

Attribute	Description
Name, age	Ion Rusu, 48 years old
Occupation	Owner of a 25 ha vegetable farm in Criuleni district (tomatoes, peppers, cabbage, onions)
Sales today	About 70% of the harvest is sold wholesale: to two regular buyers, to resellers at the wholesale market and through ads on 999.md
Digital skills	Uses a smartphone every day (Viber, Facebook, 999.md), uses a computer rarely
Goals	Sell the harvest quickly during the peak season, get a fair price, keep regular buyers from year to year
Frustrations	Spends hours on calls, buyers cancel at the last moment, intermediaries push prices down, has to remember who promised what
What would help	Publishing an offer from the phone in a few minutes, seeing all orders in one place, being notified when someone orders

Ion is not against technology, but he will not spend an evening learning a new system during the harvest. For him a platform is useful only if it saves time from the very first offer he publishes.

The second persona, in Table 2.5, is a buyer who needs regular deliveries and spends a large part of the working day coordinating suppliers.

Table 2.5 – Persona 2: the distributor (wholesale buyer)

Attribute	Description
Name, age	Natalia Ceban, 35 years old
Occupation	Procurement manager at a regional distribution company that supplies about 40 shops and restaurants in Chișinău
Purchases today	Buys fresh vegetables and fruit weekly from 10–15 producers, mostly by phone, and keeps orders in an Excel file
Digital skills	Works on a laptop and a smartphone all day, is used to online services and e-mail
Goals	Stable supply of the needed quantity and quality, predictable prices, fewer phone calls
Frustrations	Does not know what is actually available until calling, has to call several farmers to complete one order, has no history to compare suppliers
What would help	Search by product and region, real stock on every listing, order history and reviews of suppliers

The needs of the two personas are complementary: Ion wants to spend less time looking for buyers, and Natalia less time looking for producers. Both would accept a new tool only if it is faster than a phone call, which is the main design constraint of the platform.

2.4 Simplified SWOT analysis

The SWOT analysis in Table 2.6 sums up the position of the proposed platform relative to the competitors. Strengths and weaknesses refer to the platform itself; opportunities and threats describe the environment.

Table 2.6 – Simplified SWOT analysis of the proposed platform

Strengths	Weaknesses
- focus on wholesale trade in farm produce for the local market; - interface in Romanian, works in the browser of any phone; - stock is linked to orders and updated automatically; - each order has its own status and conversation; - verification of accounts and an automatic check of new listings; - no fees for producers at the launch stage.	- no users at launch, so the platform is not useful until both sides join; - limited development resources, a single developer; - no integrated payments, invoicing or transport; - web application only, no native mobile application; - depends on the digital skills of older farmers.
Opportunities	**Threats**
- the EU integration process, which raises the requirements for traceability [9]; - the growing acceptance of digital channels in B2B trade [10]; - partnerships with farmer associations and cooperatives; - future modules for payments, electronic invoices and route planning; - extension towards exports and regional markets.	- strong habit of solving everything by phone; - 999.md or Agromag.md may add similar B2B functions; - strong seasonal and weather-related fluctuations of supply [11]; - risk of fraud or unfair reviews, which can damage trust; - unwillingness of some intermediaries to lose their role.

The main risk is not technical but the so-called “cold start”: a marketplace is useful to buyers only if it already has sellers, and the other way around. For this reason the platform must be useful to a single producer from the first day, and the objectives in Chapter 3 are set for a pilot group of real users.

3 PURPOSE AND SMART OBJECTIVES

The initial purpose of the project, formulated in the first project proposal, was the digitalisation and streamlining of wholesale trade between agricultural producers and distributors through a dedicated web platform for managing offers, real-time stock and direct orders. The research in Chapter 2 confirmed this direction, but it also showed what matters most to the users: the time lost on calls, the trust between unknown partners and the risk of selling the same lot twice.

3.1 Refined purpose of the project

Taking these conclusions into account, the purpose was refined so that it answers directly the question “why is this application useful?”:

> *“To let Moldovan farmers and wholesale buyers agree and follow wholesale deals in minutes instead of hours of phone calls, by keeping offers, real stock, orders and the conversation about them in one place, between verified participants.”*

Compared to the initial version, the refined purpose names the users, states the benefit in terms of time and adds trust, which the research identified as a condition for working with new partners. It also stays within the scope of the project: payments, invoicing and transport remain outside it.

3.2 SMART objectives

Following the SMART criteria, each objective is specific, measurable, achievable, relevant and time-bound [12]. Both are measured during a pilot operation of the platform between 1 February and 31 May 2027, with at least 10 producers and 5 distributors from Moldova. The first objective concerns the saving of time, the second one the experience of the users.

Objective 1 (efficiency): during the pilot operation, the median time between a distributor placing an order and the seller's decision to accept or reject it is under 4 hours, measured on at least 30 orders. Table 3.1 shows how the objective meets each SMART criterion.

Table 3.1 – SMART analysis of Objective 1

Criterion	How the objective meets it
Specific	It concerns one step of the deal, the answer of the seller to an order, which today depends on phone calls
Measurable	The time is calculated from the moments of placing the order and of the decision, which the platform already stores for every order and notification
Achievable	The seller is notified at once in the application and by e-mail, and can accept or reject the order with one action from the phone
Relevant	It answers the key problem of the slow search for a partner and protects perishable produce from waiting
Time-bound	Measured between 1 February and 31 May 2027, on at least 30 orders

The value of 4 hours was chosen because a decision taken within the same half of the working day allows the goods to be delivered the same or the next day, which is essential for fresh vegetables and fruit. The result will also be compared with the time that the respondents report for arranging a deal by phone.

Objective 2 (user experience): during the pilot operation, the platform reaches an average System Usability Scale (SUS) score of at least 70 among at least 15 pilot users, and at least 80% of the new sellers publish their first listing in under 3 minutes without help. Table 3.2 shows the SMART analysis.

Table 3.2 – SMART analysis of Objective 2

Criterion	How the objective meets it
Specific	It concerns the ease of use of the platform for its two primary user groups
Measurable	SUS is a standard ten-question questionnaire with a score from 0 to 100 [13]; the time to publish a listing is measured during the first use
Achievable	Publishing a listing requires one form, the interface is in Romanian and works in the phone browser, without installation
Relevant	Producers will leave the phone for a platform only if it is faster and simpler, as shown by the personas
Time-bound	Measured at the end of the pilot operation, by 31 May 2027

A SUS score of 70 is above the average of about 68 reported for a large number of studied systems [14], so the objective asks for a platform that is better than average, not perfect. Together, the two objectives check both sides of the value of the platform: that it saves time and that people are willing to use it.

4 TARGET AUDIENCE AND UNIQUE VALUE PROPOSITION

This chapter defines for whom the application is built and why these people would choose it instead of the phone, a classified ads website or another platform. Market segmentation divides a market into groups of customers with similar needs and behaviour, using geographic, demographic, psychographic and behavioural criteria [15].

4.1 Target audience segments

Four segments were identified: two primary ones, which generate the transactions, and two secondary ones, which support the functioning of the platform. They are described in Table 4.1.

Table 4.1 – Target audience segments

Segment	Type	Who they are	Demographic characteristics	Psychographic characteristics
S1. Agricultural producers (sellers)	Primary	Small and medium farms, peasant farms and agricultural enterprises that sell vegetables, fruit or grain wholesale	Mostly 35–60 years old, rural areas, owners or family members who run the farm; lots from a few hundred kilograms to tens of tonnes	Pragmatic, short of time in the harvest season, rely on personal relationships, cautious about fees and new tools, prefer the phone to the computer
S2. Wholesale buyers (distributors)	Primary	Distribution companies, small retail chains, HoReCa businesses and processors that buy regularly	Mostly 25–45 years old, procurement managers or owners in Chișinău, Bălți and district centres; buy weekly or more often	Oriented to efficiency, used to online services, value reliability and predictability more than the lowest price, need a documented history
S3. Platform administrators	Secondary	Platform staff who verify accounts and listings and handle disputes	One to three people at launch, good digital skills	Responsible for the trust in the platform, need fast moderation tools and clear rules
S4. Partners	Secondary	Farmer associations and cooperatives, transport companies, agricultural consultants	Organisations that already have networks of producers or clients	Interested in the digitalisation of their members, can bring the first users and future integrations

The primary segments are very different from each other. Producers are less digital and more sensitive to costs, while buyers are used to online tools but have little time for anything that does not bring results quickly. The secondary segments do not trade, but administrators keep the catalogue trustworthy and partners such as farmer associations can bring many producers at once.

4.2 Key problems addressed by the application

The problems of the current trade process were grouped by their effect on the two primary segments. Three key problems remain, and they define the core of the platform. They are presented in Table 4.2.

Table 4.2 – Key problems solved by the platform

Key problem	Affected segments	How it shows today	How the platform solves it
KP1. Finding a reliable partner takes too long	S1, S2	Hours of calls and waiting for answers to ads; deals limited to a small circle of known partners	Catalogue of listings with search by product and region, verified accounts and public reviews after completed orders
KP2. Stock and agreements are not recorded in one place	S1, S2	Lots promised twice, oral agreements, disputes about quantity and price	Each listing shows the real stock, which decreases when an order is confirmed; each order keeps its price, quantity and status
KP3. Communication around an order is slow and scattered	S1, S2, S3	Details lost between calls, Viber and Facebook; perishable produce waits for an answer	Messages attached to each order, e-mail and in-app notifications, a personal dashboard with all orders

The three problems are connected. Without trust a buyer does not risk ordering from an unknown producer; without a shared record even a trusted partner can disappoint; and without fast communication perishable goods lose value while the parties wait for each other.

4.3 Unique value proposition

A value proposition describes how a product relieves the pains of its customers and creates the gains they expect [16]. The unique value proposition of AgroHub is:

> *“A free, Romanian-language B2B marketplace where Moldovan farmers and wholesale buyers trade agricultural produce directly: every listing shows the stock that is really available, every deal becomes an order with a clear status and its own conversation, and every participant is verified, so a harvest can be sold with a few taps on a phone instead of a day of phone calls.”*

Verified participants and public reviews answer the first key problem, the real stock and the order with a status answer the second, and the conversation attached to each order, with notifications, answers the third. In a short form, the UVP becomes “Sell your harvest, not your time on the phone” for producers and “Order from verified farmers, with real stock and no calls” for buyers.

Table 4.3 compares the main advantages of the platform with the three groups of alternatives that the target audience uses today.

Table 4.3 – Main competitive advantages compared to the alternatives

Advantage	Compared to the phone and messengers	Compared to 999.md	Compared to specialised platforms
Built for wholesale trade in farm produce	Reach beyond personal contacts	Each listing has quantity, price per kilogram, region, harvest date and delivery terms instead of free text	Local scale and lots of a few tonnes, not export containers or farm inputs
Real stock linked to orders	Nothing has to be remembered, the stock updates itself	An ad does not show what is left	Found elsewhere mostly in paid enterprise software
Order with a status and its own conversation	Agreements are written, not oral	The deal continues on the platform, not by phone	Similar functions, but free and in Romanian
Verification and reviews	Trust beyond the circle of acquaintances	Accounts and listings are checked, including an automatic check of every new listing	Similar level of trust, adapted to small producers
No fees, works in the browser of any phone	No extra costs, nothing to install	The same accessibility, with more structure	Not available in Moldova or paid

What makes the platform different is the combination of these advantages and its focus on one job: wholesale trade in fresh produce between Moldovan farmers and the companies that buy from them. This focus is its main competitive advantage.

5 CONCLUSIONS

The research shows that the proposed application addresses a real need. Wholesale trade in agricultural products is important for the economy of Moldova, yet it is still organised by phone and in messenger groups, with time lost, lots promised twice and little trust between unknown partners. The five analysed systems each cover part of the problem, but none combines a local, free catalogue with real stock, structured orders and verified participants.

The target audience is clearly defined: small and medium producers who sell wholesale and the distributors, shops and HoReCa businesses that buy from them, supported by administrators and partner organisations. The refined purpose and the two SMART objectives turn the idea into measurable results, a median answer time under 4 hours and a SUS score of at least 70, both to be verified in the pilot operation by 31 May 2027. The unique value proposition explains why users would choose the platform: it lets them sell and buy produce faster than by phone, with a real stock and partners they can trust.

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